Lab-2: Measuring the *Hubble* Diagram using Standard Rulers

1 Introduction

In 1929, Edwin Hubble measured that distant galaxies were systematically redshifted relative to galaxies that were closer (see Figure 1). From this data, Hubble inferred that the universe was expanding, an idea initially worked out by Georges Lemaitre using Einstein's theory of gravity. In this lab, you will conduct a measurement similar to Hubble's and will produce your own version of his famous Hubble diagram shown below.

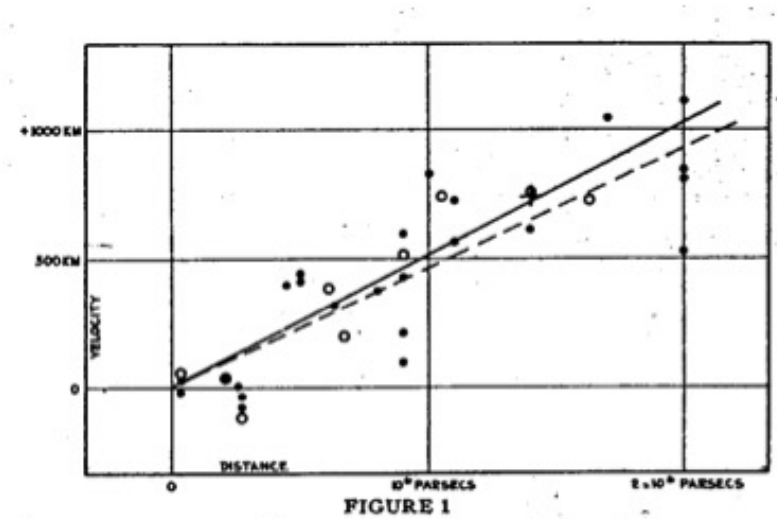

Figure 1: Velocity-Distance Relation among Extra-Galactic Nebulae (Hubble et al. 1929).

2 Method

In Figure 2, we illustrate the impact of an expanding universe of photons emitted from distant objects. Because the speed of light is constant, photons that we measure today were emitted in the past, with photons originating from objects that are further away being emitted earlier in time. This means that photons from objects that are further away are older, and thus, those photons have experienced more expansion by the universe. This plot below illustrates the relationship between the expansion experienced by a photon and the distance of its emitter.

From this relationship, we can determine whether the universe is expanding, contracting, or static by looking at a number of galaxies and measuring their distance (corresponding to the x-axis) and the expansion experienced by their photons (corresponding to the y-axis).

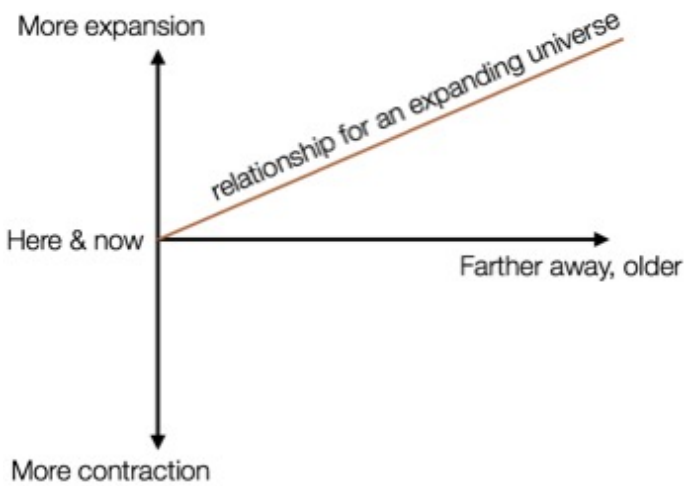

Figure 2:

2.1 Measuring Distances

We will use geometry to measure the distance of our galaxies, by assuming that they can be used as *standard rulers*, objects with the same physical size. Galaxies that are closer will look bigger and subtend a larger angle. Whereas galaxies that are further away will look smaller and subtend a smaller angle. This relationship between the angular size of the galaxy and its distance is illustrated in Figure 3.

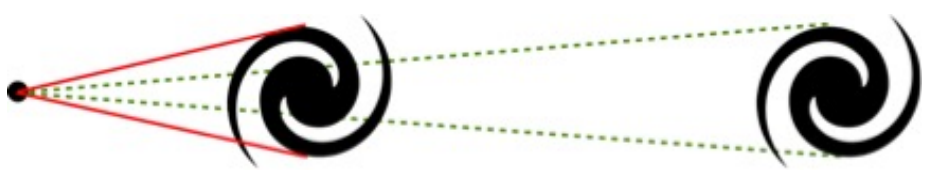

Figure 3:

Typical galaxies are about 22 kpc across. Using the geometry shown in the Figure 3, we can arrive at the following relationship:

$$\theta_{gal} = \frac{22\,\mathrm{kpc}}{d_{gal}} \tag{1}$$

where θ_{gal} is the angular size of our galaxy images in *radians* and d_{gal} is the distance to galaxies in *kpc*. So, by measuring the angular size of our galaxy images, we can use the above equation to determine the distance to the galaxy.

3 Data and Observations

3.1 Stone Edge Observatory

The Stone Edge observatory (SEO) is a 24 inch reflecting telescope, located in Sonoma, California. Stone Edge has a charge coupled device (CCD) camera and is equipped with a full set of filters, including the same *ugriz* bands as used by the Sloan Digital Sky Survey (SDSS) (see Figure 4).

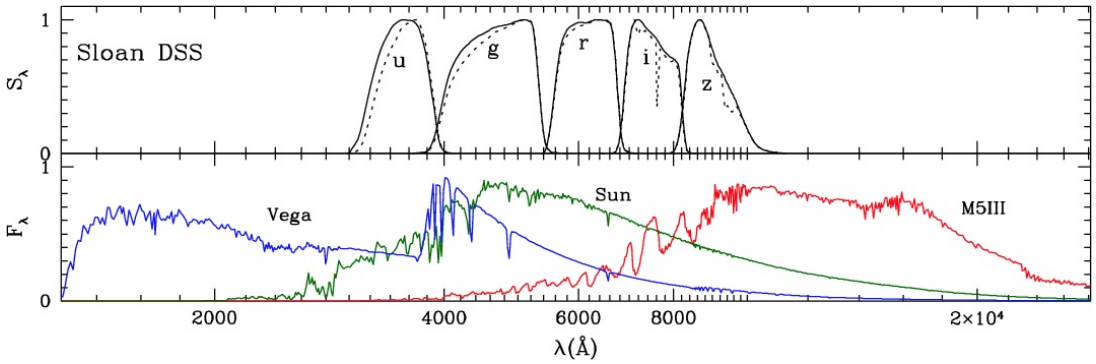

Figure 4: Filter transmission functions for SDSS, plotted above spectra for three different stars of different surface temperatures: Vega (A-type, ∼10,000 K), our Sun (G-type, ∼6,000 K), and an M5III star (M-type, < 3500 K). Its clear from this image that these three objects will look much different in our image depending on what filter bands we use. Image source: http://www.aanda.org/articles/aa/full/2004/28/aa0250-04/img19.gif

3.2 Observations

From Table 3.2, randomly select at least **six galaxies** to be observed. For each galaxy, you will likely have to look up information like: Right Ascension (RA), Declination (Dec), and redshift (z), and make sure that it is observable by SEO during the next couple of weeks. Feel free to add additional galaxies, but be mindful of its night time visibility from SEO, and also its magnitude and distance. In practice, to have reasonably modest observations, in terms of exposure time and angular extent, I would guess that a reasonable magnitude (M) and redshift range to be something like $15 < M < 6$ and $0.002 < z < 0.1$, respectively. Be careful to not select too many galaxies near the low end of that redshift range, as the gravitational pull from nearby large scale structures can contribute peculiar velocities of ∼300 km/s.

Galaxy Name	z	R.A. (J2000)	Decl. (J2000)	Notes
NGC 2775				
NGC 3034				
NGC 3147	0.0093	10:16:53.7	73:24:03	
NGC 3227				
NGC 3368				
NGC 3516				
NGC 3941				
NGC 4486				
NGC 4501	0.0076	12:31:59.2	14:25:12	
NGC 4565	0.0041	12:36:20.8	25:59:16	
NGC 4579				
NGC 4775				
NGC 5248				
NGC 5548	0.0172	14:17:59.5	25:08:12	
NGC 5746				
NGC 5907				
NGC 6015				
NGC 6181	0.0079	16:32:20.9	19:49:36	
NGC 6217				
NGC 6643				
NGC 6764				
NGC 7331				

Table 1: A list of potential galaxies that should be reasonable to observe for this lab, with space given for the redshift, Right Ascension (R.A.), and Declination (Decl.) for each object. You should choose a mix of spiral and elliptical galaxies.

Figure 5: Two galaxies that you might observe for this lab. Both images are *grz* image taken from SDSS (`https://skyserver.sdss.org/`), and are 1024 by 1024 pixels across, where each pixel is 0.8-arcsec resolution. (Left) NGC 4565 is a edge-on spiral galaxy about 40 million light years away. (Right) NGC 6181 is a barred spiral galaxy about 100 million light years away.

3.3 Instructions for Scheduling Observations

As before, for real-time observing time on SEO, following these instructions:

1. Reserve time on the SEO at `https://voices.uchicago.edu/stoneedgeobservatory/`.
2. To observe, use the SEO Slack and the **itzamna** channel. For observing instructions, use the SEO cheat sheet at `https://github.com/bradfordbenson/ASTR21200_2026/blob/main/Labs/ASTR21200_SEO_Cheat_Sheet.pdf`
3. If in doubt, also refer to the instructions from the SEO description in Lecture-4 [1]

4 Data Analysis

4.1 Making RGB Images

In this section, you will make red-green-blue (RGB) color images of each galaxy that you observed last week. Download your data from `stars.uchicago.edu`. Make an RGB image

[1] `https://github.com/bradfordbenson/ASTR21200_2026/blob/main/Lectures/2026_ASTR21200_Lecture4_Intro_to_SEO.pdf`

as you did in the last lab, making a composite image from your *gri* filter measurements, with i is your reddest color and g is your bluest color. Its recommended you use the ds9 software `https://sites.google.com/cfa.harvard.edu/saoimageds9/download`, but other options include python [2]; or the ASTAP software package [3]. If you were to take multiple images that you wanted to *coadd*, to make a higher signal-to-noise final image, you can use the *Swarp* program which is described more on the class wiki at `https://github.com/bradfordbenson/ASTR21200_2026/wiki/Coadd-Images`.

When you take your images, you should assess how they look, i.e., do you clearly see each galaxy? If you are disappointed they could have been affected by bad weather, and might also be improved with longer exposures. Feel free to take additional observations, either repeating the same observations (i.e., hoping for better weather, or to coadd more observations for better signal-to-noise), or increasing your exposure time. For example, if you took 30-sec exposures and it looks too faint, maybe try 120-sec or 240-sec exposures this time. Also consider trying to*coadd* multiple observations (see instructions above).

4.2 Measuring the Angular Size of Galaxies

In this section, we will measure each galaxy's angular size. You can use whatever program you want, but ds9 or python should be the most straightforward. It shouldn't matter significantly which filter band you use, though remember your bluer bands will be more sensitive, and you might want to check and comment on any differences.

For your lab report, you will want to include the following information:

1. Which galaxies did you observe? You will want to include a Table with key properties of your galaxies, e.g., R.A., Decl., redshift, and measured angular size.
2. From your observations, can you tell what kind of galaxy is it (e.g., spiral, elliptical, unclear)?
3. What is the angular size of your galaxy? How did you define the angular extent of the galaxies? Did you use a uniform threshold across your images?
4. Given that the galaxies have relatively low surface brightness, did you have a hard time measuring the extent of your galaxy? Can you estimate the level of systematic uncertainty in your measurement? Are there any challenges due to the orientation of the galaxies in your image? Are there any differences between the *gri* images? Can you think of other sources of systematic uncertainty in the way you did your measurement?

[2] `https://docs.astropy.org/en/stable/visualization/rgb.html`
[3] `https://www.hnsky.org/astap.htm`

4.3 Compare to SDSS or Legacy Survey Images

There are some systematic uncertainties that could be intrinsic to our method or assumptions. However, there could also be some systematic uncertainties that could be better controlled with higher sensitivity data (e.g., if the SEO sky brightness or depth was causing a noise bias in our size measurements). To try and control for this latter type of systematic uncertainty, for at least two of your galaxies download the fits data from either the Legacy Survey (LS) or SDSS image server. As a reminder from Homework-3, these are the instructions:

1. For LegacySurvey, you can use a call like this: `https://www.legacysurvey.org/viewer/fits-cutout?ra=187.7059&dec=12.3911&width=1024&height=1024&pixscale=0.262&layer=ls-dr10&bands=gri`
2. For SDSS, to download the fits files, its probably easiest to use python, an example is on the wiki at: `https://github.com/bradfordbenson/ASTR21200_2026/wiki/SDSS-Legacy-Survey`

For the LS or SDSS fits images, repeat your angular size measurements as you did them on the SEO data. How do your measurements of angular size compare to the SEO measurements? How different are the measurements of angular size between SEO and LS or SDSS? If they are significantly different sizes, can you tell what effect is causing it? For example, is it something due to the depth of the SEO data? And how is it affecting your angular size measurement? Use any difference to help estimate a systematic uncertainty in your measurement due to the depth of the SEO data. Given what you've learned, can you think any improvements to your original methodology using the SEO data? Or is there a way that you can systematically correct your SEO measurements for the effect you've found comparing to LS or SDSS?

4.4 Measuring the Expansion Rate of the Universe, or the Hubble Constant

The wavelength of light changes as the universe expands, an effect known as cosmological redshift. If the universe expands, the wavelength is stretched, becoming longer and redder. For a contracting universe, the wavelength will be compressed becoming bluer. We define the redshift, z, as:

$$z = \frac{(\lambda_{measured} - \lambda_{original})}{\lambda_{original}} \tag{2}$$

Where $\lambda_{measured}$ is the wavelength of the light measured at Earth, and $\lambda_{original}$ is the wavelength of the light when it was originally emitted at the source. Therefore the redshift, z,

is a measure of how much the wavelength of the light has been stretched or compressed, due to the relative motion of Earth relative to the source galaxy. Typically redshift is measured using spectroscopy, where the location of well-known emission and absorption lines of particular atoms or molecules are measured, then compared to their known laboratory-measured values, and then giving a measurement of the effectively recessional velocity of your galaxy, or redshift. Using Simbad or NED, find the redshift for each galaxy that you observed.

For reporting your *Hubble* diagram and constant measurements, you should include the following information in your lab report:

1. Using your measurements, make a plot of distance versus redshift for your galaxies, and include it in your lab report. Make sure to include error bars in your distance measurements.

2. By using the relationship between redshift and velocity as $v = c\,z$, where c is the speed of light, also make a plot of distance versus velocity. Use this measurement to estimate a best-fit *Hubble* constant, which should be in units of (km / sec / Mpc). Also estimate the uncertainty in your best-fit *Hubble* constant measurement.

5 Discussion

Discuss your results, and the level of agreement with previous results. Discuss any issues or systematic uncertainties that you've identified during your analysis, and how they affected your estimate of the *Hubble* constant; be quantitative. You should include answers to the questions below in your discussion:

1. Is your best-fit *Hubble* constant a good fit to your measurement? What is the χ^2 to the fit? Can you estimate a probability to exceed (PTE) value of the fit of the model to your data?

2. How does your *Hubble* constant measurement compare to the accepted value of 70 (km / sec / Mpc)? Does it agree within the level of uncertainty (σ) of your *Hubble* constant measurement? How many σ off is the accepted value from your best-fit value?

3. Discuss differences between the statistical and systematic uncertainties that effect this measurement. For example, if we measured 100 more galaxies this way, the statistical uncertainty should decrease like $\sqrt{N_{galaxies}}$, so could we get an arbitrarily tight constraint on the Hubble constant this way? If not, what systematics limit this measurement? And how do the systematics described in Section 4.2 and 4.3, affect your measurement here?

6 Lab Report

Prepare a *jupyter* notebook that documents your entire analysis for the lab. Make sure to explain your steps and conclusions; imagine writing a tutorial for another astronomy student, who is not taking the class. Use *markdown* boxes (which can also parse LaTeX). Note that you can also include figures (i.e., in png, jpg, etc. form) that are produced outside of the notebook (e.g. with ds9).

The explanations in the *jupyter* notebook will be what we read, but we might look at your code if we think you did something wrong. Make sure that the report is logical; each section should have a short introduction, then code with results and plots, then a conclusion. Make sure the section numbering follows this manual (e.g., Introduction, Data, Data Analysis, Conclusions). Once your notebook is finished, make sure to restart it and re-run all cells. Then save the notebook in pdf format, e.g., through the print menu.

A Using ds9 to measure angular extent

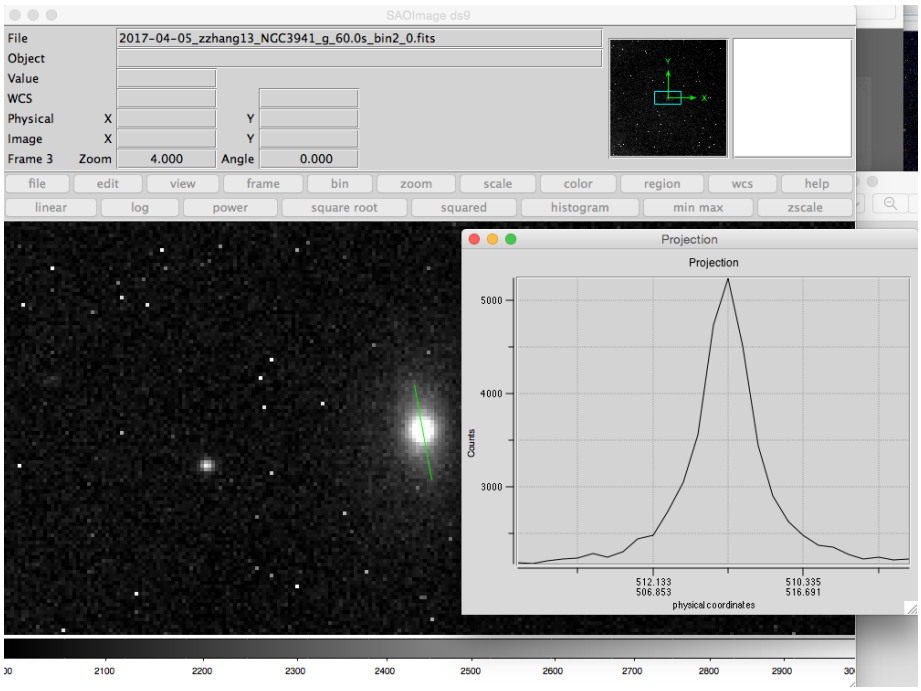

Figure 6: Using ds9 to measure the angular size of a galaxy. In this image, the green line shows the approximate extent of the galaxy, and the projection plot shows the photon counts measured in this cut through the galaxy.

Under the *Scale* tab, click on *Log*. Also under the *Scale* tab, go to *Scale Parameters* which will open a plot of the *Pixel Distribution*, a histogram of the number of counts per pixel in your image. Feel free to play around, but I would recommend choosing *Low* and *High* values above the minimum and significantly below the maximum. For my images, *Low* and *High* values of ∼2000 and 3000, respectively, worked well. For this step, you might want to *zoom* into your galaxy to better see how the values affect how your galaxy looks.

Next, *zoom* in on your galaxy to measure its angular extent. From the *Region* tab, click on *Region* > *Shape* > *Projection*. Click and drag your cursor across the source (see Figure 6). A new window will open displaying the brightness profile of the image along the line you drew. Measure how many pixels across your galaxy is using the end-points of the line that you drew. The Stone Edge Observatory pixel scale is 1.52 arc seconds per pixel if you used a setting of bin=2. Convert your measurement to arc seconds.

For more information about using ds9, you can use this helpful tutorial:

https://astrobites.org/2011/03/09/how-to-use-sao-ds9-to-examine-astronomical-images/